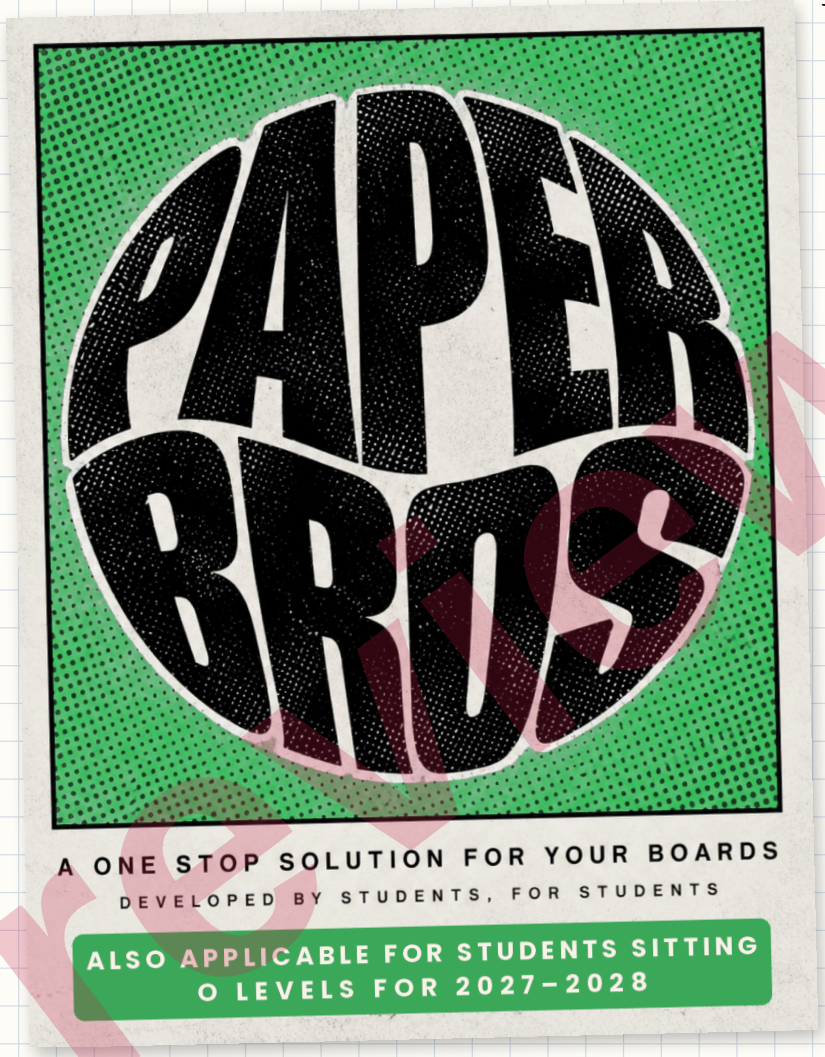
PREVIEW

Cambridge O Level

Biology 5090

the complete notes — for exams in 2026, 2027 & 2028



notes for every syllabus point · labelled diagrams
worked examples & exam tricks · Paper 1 & Paper 2 practice + answers

this notebook belongs to:

Name .....

School / class ..... Exam: June / Nov

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what's inside

Contents

tick the box when a topic is done & dusted ✓

#	Topic	What's inside (syllabus sections)	Notes pg	Practice pg	✓
–	The exam at a glance	papers, marks, AOs, command words, how to use these notes	3	–	☐
–	Definitions & practical toolkit	the definitions to learn word-for-word; drawings, tables, graphs, variables, tests	4	–	☐
1	Cells	1.1 cell structure & function · 1.2 specialised cells, tissues, organs, magnification	6	9	☐
2	Classification	2.1 species, binomial names, keys · 2.2 five kingdoms, vertebrates, arthropods, plants, viruses	11	14	☐
3	Movement into & out of cells	3.1 diffusion, osmosis, turgor & plasmolysis · 3.2 active transport	16	19	☐
4	Biological molecules	4.1 elements, building blocks, food tests	21	23	☐
5	Enzymes	5.1 lock & key · 5.2 effects of temperature and pH	25	28	☐
6	Plant nutrition	6.1 photosynthesis, limiting factors · 6.2 leaf structure · 6.3 nitrate & magnesium	30	34	☐
7	Transport in flowering plants	7.1 root hairs, water pathway · 7.2 transpiration, translocation, xylem	37	40	☐
8	Human nutrition	8.1 diet · 8.2 digestive system, teeth, enzymes, bile · 8.3 villi, absorption	43	47	☐
9	Human gas exchange	9.1 lungs, alveoli, breathing, air composition, cilia & mucus	50	54	☐
10	Respiration	10.1 uses of energy · 10.2 aerobic · 10.3 anaerobic, oxygen debt	57	60	☐
11	Transport in humans	11.1 double circulation · 11.2 heart, CHD · 11.3 blood vessels · 11.4 blood	63	68	☐
12	Disease & immunity	12.1 pathogens, malaria, HIV, cholera, alcohol, smoking · 12.2 antibiotics · 12.3 immunity	71	77	☐
13	Excretion	13.1 waste products · 13.2 kidneys, nephron, deamination	79	82	☐
14	Coordination & control	14.1 nerves & synapses · 14.2 the eye · 14.3 hormones · 14.4–14.6 homeostasis	84	90	☐
15	Coordination in plants	15.1 tropisms and auxin	93	95	☐
16	Continuity of life	16.1 mitosis & meiosis · 16.2 sexual vs asexual · 16.3 flowers · 16.4 human reproduction	96	103	☐
17	Inheritance	17.1 variation · 17.2 DNA · 17.3 genetic crosses · 17.4 natural & artificial selection	106	110	☐
18	Biotechnology & GM	18.1 yeast, bacteria, fermenters, enzymes · 18.2 genetic modification	113	116	☐
19	Organisms & environment	19.1 energy flow · 19.2 nutrient cycles · 19.3 ecosystems · 19.4 human impact · 19.5 conservation	117	122	☐
✶	Answers	Paper 1 keys + Paper 2 mark-scheme style answers for every topic	125	–	☐

Valid for 2026, 2027 & 2028 ✔

This book follows the 5090 syllabus for exams in **2026, 2027 and 2028** (version 4). Cambridge says there are **no significant changes which affect teaching**, and textbooks endorsed from 2023 are still fine.

How I'd use this book

1. read the notes → 2. cover the labels & redraw the diagrams → 3. do the practice without peeking → 4. mark with the answers at the back & fix your mistakes in red.

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before we start...

The Exam at a Glance

You sit **three** papers. Everyone takes Paper 1 **and** Paper 2, then **either** Paper 3 **or** Paper 4. Grades go from **A*** down to **E**.

Paper	Time	Marks	Weight	What it looks like	Tests
1 Multiple Choice	1 h	40	30%	40 compulsory four-option (A–D) questions	AO1 + AO2
2 Theory	1 h 45 min	80	50%	compulsory short-answer & structured questions	AO1 + AO2
3 Practical Test	1 h 30 min	40	20%	you do real experiments in a lab	AO3
		– OR –			
4 Alternative to Practical	1 h	40	20%	written questions about experiments (no lab)	AO3

AO Assessment objectives

Objective	Weight
AO1 Knowledge with understanding – facts, definitions, vocabulary, apparatus	50%
AO2 Handling information & problem-solving – data, graphs, calculations, new situations	30%
AO3 Experimental skills & investigations – planning, observing, evaluating	20%

Papers 1 & 2 are about **63% AO1 : 37% AO2**. So a big chunk of the marks is "apply what you know to something you haven't seen" – graphs, tables and unfamiliar organisms.

biology = precise words ⚠

Examiners mark **key words**. "Water moves" earns nothing; "water moves by **osmosis** from a **higher water potential** to a **lower water potential** through a **partially permeable membrane**" earns the lot.

key How to read these notes

- **maroon bold** = a key word that earns the mark
- yellow highlight = learn word-for-word
- yellow box = **definition**; blue box = **worked example**
- sticky notes = exam tips & traps ⚠
- little grey numbers like [8.2.9] = the syllabus point
- every diagram is drawn the way you should draw it in the exam: clear lines, ruled labels

✱ Units you'll meet

length	volume	mass	energy
1 mm = 1000 µm	1 dm^3 = 1000 cm^3	1 g = 1000 mg	1 kJ = 1000 J
1 cm = 10 mm	1 cm^3 = 1 ml	1 kg = 1000 g	(kJ per gram of food)

Top 3 mark-losers

① **describing** when asked to **explain** (or the other way round) ② vague words: "it", "stuff", "goes up" ③ forgetting **units** when quoting data from a graph.

cmd Command words

The first word of the question tells you what the examiner wants:

Word	What you do
State / Give	short clear answer, no explanation needed
Define	the precise meaning – learn definitions word-for-word
Describe	the main features or what happens; for a graph: trend + figures
Explain	give reasons *why* or *how* – use "because..." and link the steps
Outline	set out the main points, briefly
Compare	similarities and/or differences – mention **both** things
Contrast	differences only
Discuss	write about the issue in depth: points for and against
Evaluate / Assess	judge the quality or importance – give a conclusion
Calculate	work it out – show working, answer + unit
Determine	get an answer using the data / graph given
Suggest	apply your biology to a new situation – several answers can be right
Predict	say what may happen using the information
Identify	name / select / recognise
Comment / Justify	give an informed opinion / support a case with evidence
Sketch	simple freehand drawing showing the key features, proportions right

✱ Describing a graph: the 3-step method

WE ✎ "Describe the effect of temperature on enzyme activity"

① **overall trend**: activity increases as temperature rises up to 40 °C,
② **change of trend**: then decreases rapidly above 40 °C,
③ **quote data with units**: from 12 units at 20 °C to a maximum of 30 units at 40 °C, and zero at 60 °C

Preview

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learn these word-for-word

Definitions You Must Know

cover the right-hand column & test yourself every week

term	definition
cells, classification & transport in cells	
species	a group of organisms that can reproduce to produce **fertile offspring**
binomial system	internationally agreed naming system in which the scientific name has two parts: **genus** and **species**
diffusion	the net movement of molecules or ions from a region of their **higher concentration** to a region of their **lower concentration** (down a concentration gradient), as a result of their **random movement**
osmosis	the net movement of **water molecules** from a region of **higher water potential** to a region of **lower water potential**, through a **partially permeable membrane**
active transport	movement of molecules or ions through the cell membrane from **lower** to **higher** concentration (against a concentration gradient), using **energy released during respiration**
catalyst	a substance that increases the rate of a chemical reaction and is **not changed** by the reaction
enzymes	**proteins** that function as **biological catalysts**
plants & nutrition	
photosynthesis	the process by which plants make **carbohydrates** from raw materials using **energy from light**
transpiration	loss of **water vapour** from leaves
translocation	movement of **sucrose and amino acids** in the **phloem** from **sources** to **sinks**
physical digestion	breakdown of food into smaller pieces **without chemical change** to the food molecules
chemical digestion	breakdown of **large molecules** into **small molecules**
absorption	movement of nutrients from the intestines into the cells lining the digestive system and then into the **blood**
assimilation	uptake and **use by cells** of nutrients from the blood
peristalsis	waves of contractions of **longitudinal** and **circular muscles** which move food through the gut
respiration, transport, disease	
respiration	the chemical reactions in **all living cells** that **release energy from glucose**
aerobic respiration	release of a relatively **large** amount of energy by the breakdown of glucose **in the presence of oxygen**
anaerobic respiration	release of a relatively **small** amount of energy by the breakdown of glucose **without using oxygen**
double circulation	blood passes through the heart **twice** for each complete circuit
pathogen	a **disease-causing** organism
transmissible disease	a disease in which the pathogen can be **passed from one host to another**
drug	any substance taken into the body that **modifies or affects chemical reactions** in the body
active immunity	defence against a pathogen by **antibody production in the body**
passive immunity	**short-term** defence by antibodies **acquired from another individual** (across the placenta, in breast milk)
excretion	removal of **toxic materials** and the **waste products of metabolism** from organisms
deamination	removal of the **nitrogen-containing part** of amino acids, forming **urea** (in the liver)

term	definition
coordination	
reflex action	a **rapid and automatic** response to a stimulus
synapse	a **junction** between two neurones
hormone	a **chemical substance**, produced by a **gland** and **carried by the blood**, which alters the activity of one or more specific **target organs**
homeostasis	maintenance of a **constant internal environment**
gravitropism / phototropism	a response in which parts of a plant **grow towards or away from** gravity / light
cell division & reproduction	
haploid nucleus	a nucleus containing a **single set** of chromosomes
diploid nucleus	a nucleus containing **two sets** of chromosomes
mitosis	nuclear division giving rise to **genetically identical** cells in which the chromosome number is **maintained**
meiosis	**reduction division**: chromosome number halved from diploid to haploid, giving **genetically different** cells
stem cells	**unspecialised** cells that divide by mitosis to produce daughter cells that can become **specialised**
asexual reproduction	a process resulting in **genetically identical** offspring from **one parent**
sexual reproduction	**fusion of haploid nuclei** (fertilisation) to form a **diploid zygote**, giving genetically different offspring
fertilisation	**fusion of the nuclei** from a male gamete and a female gamete
inheritance & biotechnology	
variation	**differences** between individuals of the **same species**
gene	a **length of DNA** that **codes for a protein**
allele	an **alternative form** of a gene
inheritance	transmission of **genetic information** from generation to generation
gene mutation	a **random change** in the **base sequence** of DNA
chromosome mutation	a change in chromosome **number or structure**
genetic modification	changing the genetic material of an organism by **removing, changing or inserting individual genes**
ecology	
population	a group of organisms of **one species**, living in the **same area**, at the **same time**
community	**all of the populations** of different species in an ecosystem
ecosystem	a unit containing the **community** of organisms and their **environment**, interacting together
biodiversity	the **number of different species** that live in an area
sustainable resource	one which is **produced as rapidly as it is removed** from the environment, so it does not run out

the one equation

$$\text{magnification} = \frac{\text{image size}}{\text{actual size}}$$

– same units top and bottom!

Preview

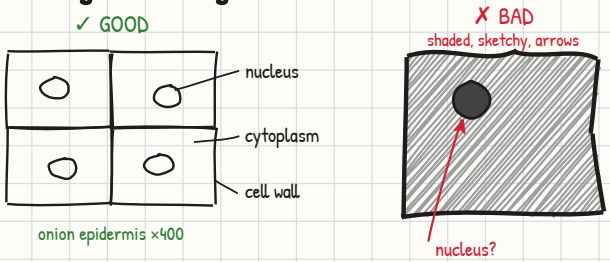
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Papers 3 & 4 made easy

The Practical Skills Toolkit

20% of your grade comes from these skills — free marks if you know the rules

✱ Biological drawings



- sharp **pencil**; **clear, continuous** lines; **no shading**
- use **most of the space** given; keep the **proportions** right
- label lines drawn with a **ruler**, **touching** the part, never crossing; no arrowheads
- draw only what you **see** — not what the textbook shows

WE 1 ✎ magnification of a drawing

A leaf is 42 mm long. Your drawing of it is 126 mm long.

magnification = image ÷ actual = 126 ÷ 42 = **×3.0** (no units; measure both in mm)

✱ Results tables

time taken for the starch to disappear

temp. / °C	trial 1 / s	trial 2 / s	trial 3 / s	mean / s
20	240	250	230	240
40	60	70	140	65

- heading = **quantity / unit**; **no units** in the body of the table
- same number of decimal places down a column
- **anomaly** (140 s) — leave it out of the mean, or repeat it: (60 + 70) ÷ 2 = 65 s

✱ Sampling with a quadrat

Place quadrats at **random** (random-number coordinates), count the plants in each, find the **mean per quadrat**, then scale up to the whole area.

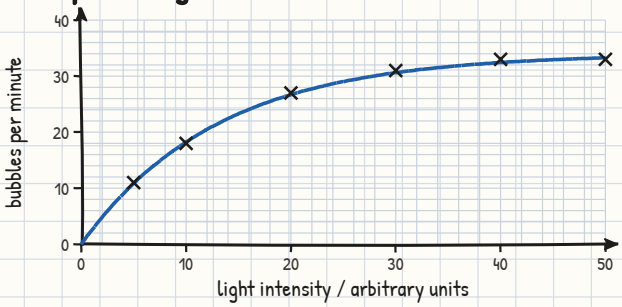
WE 2 ✎ estimating a population

10 quadrats of 0.25 m^2 held 3, 5, 0, 2, 6, 4, 1, 3, 5, 1 daisies. Field = 400 m^2.

mean = 30 ÷ 10 = 3.0 per quadrat = 3.0 ÷ 0.25 = 12 per m^2

population ≈ 12 × 400 = **4800 daisies** (range = 6 − 0 = 6)

✱ Graphs that get full marks



- **independent** variable on the **x-axis**; both axes labelled **quantity / unit**
- even scales using **more than half** the grid (steps of 1, 2, 5, 10…)
- plot points as small **crosses** (×) to within half a small square
- one thin, smooth **line of best fit** — or join point-to-point with a ruler if told to
- **bar chart** for categories (bars don't touch) · **histogram** for continuous data (bars touch) · **pie chart** for percentages

✱ Planning: the variables

independent	the one you **change** (e.g. temperature)
dependent	the one you **measure** (e.g. time for colour change)
controlled	kept the **same** (volume, concentration, pH, mass, time…) — say **how**: water bath, same syringe…
control	a set-up **without** the factor (e.g. boiled enzyme) to show the result is due to it
reliability	**repeat** ≥ 3 times and calculate a **mean**

✱ Indicators & quick tests

indicator	what it shows
hydrogencarbonate	CO_2 level: **purple** = less CO_2 than air · **red** = same as air · **yellow** = more CO_2
limewater	CO_2: colourless ⟶ **milky / cloudy**
universal indicator	pH: red (acid) ⟶ green (7) ⟶ purple (alkali)
litmus	blue ⟶ red in acid; red ⟶ blue in alkali

safety marks

eye protection always · heat Benedict's in a **water bath**, not a flame · **no flames** near ethanol · cut **away** from your fingers on a tile · wash hands after handling plants, soil or microbes.

Preview

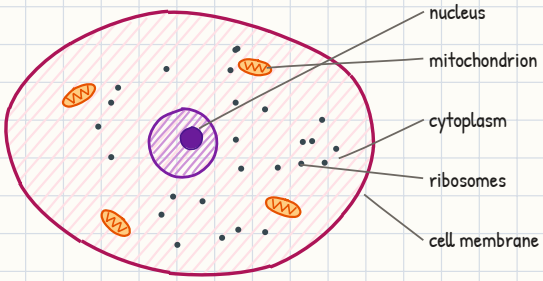
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Topic 1
Cells

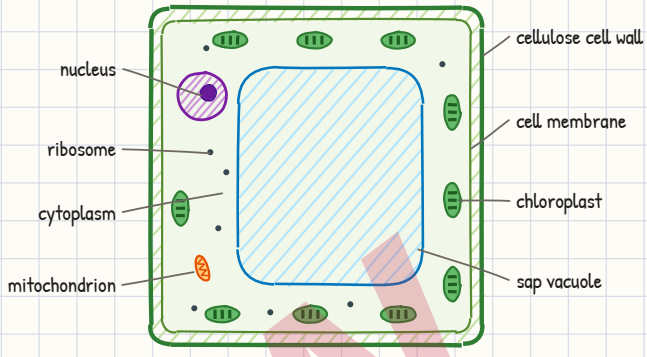
the building blocks of every living thing

1.1 Cell structure and function

Every living organism is made of one or more **cells**. A light microscope shows the nucleus, cytoplasm and membrane; tiny structures like **ribosomes** only show up clearly in **electron micrographs**. [1.1.3–4]



animal cell (e.g. a cheek cell)



plant cell (e.g. a palisade cell)

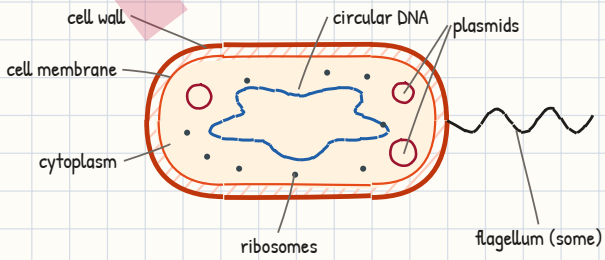
✱ What each part does [1.1.6]

structure	function	animal	plant	bacterium
cell membrane	partially permeable — **controls what enters and leaves** the cell	✓	✓	✓
cytoplasm	jelly-like; where **most chemical reactions** happen	✓	✓	✓
ribosomes	where **proteins are made** (protein synthesis)	✓	✓	✓
nucleus	contains the **chromosomes (DNA)**, which control the cell's activities	✓	✓	✗
mitochondria	site of **aerobic respiration** — release energy for the cell	✓	✓	✗
cellulose cell wall	strong and fully permeable; **supports** the cell and stops it **bursting** when it takes in water	✗	✓	wall, but not cellulose
chloroplasts	contain **chlorophyll**, which absorbs light for **photosynthesis**	✗	✓	✗
sap vacuole	stores **cell sap** (water, sugars, salts); pushes out on the wall to keep the cell **turgid**	✗	✓	✗
circular DNA	one loop of DNA carrying the genes that control the bacterium (no nucleus around it)	✗	✗	✓
plasmids	small extra **rings of DNA**; carry a few genes (e.g. antibiotic resistance); used in genetic modification	✗	✗	✓

trap ⚠ **not every plant cell has chloroplasts** — root cells live in the dark, so they have none. And animal cells **never** have a cell wall or a big permanent vacuole.

wall vs membrane the wall is **fully permeable** (lets everything through) and gives strength; the membrane is **partially permeable** and does the controlling.

✱ The bacterial cell [1.1.5]



Bacteria are **prokaryotes**: single cells that are **much smaller** than animal or plant cells (about 1–5 μm long) and have no nucleus — their DNA is a **single circular loop** lying free in the cytoplasm, plus small **plasmids**.

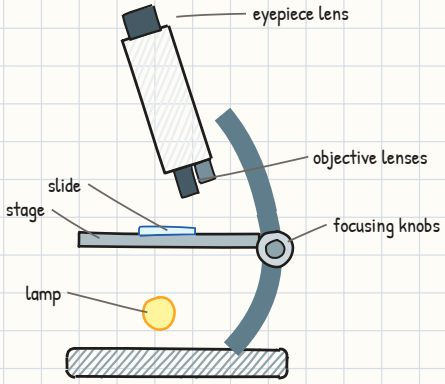
- no mitochondria, no chloroplasts, no sap vacuole
- cell wall is **not** made of cellulose
- they still have **ribosomes** (smaller ones) to make proteins

Spot the cell in 3 questions 🕵

① **Nucleus?** No ⇒ **bacterium**. ② Yes — **cell wall**, **large sap vacuole** or **chloroplasts**? Yes ⇒ **plant cell**. ③ No wall, no vacuole, irregular shape ⇒ **animal cell**.

Preview

✱ Looking at cells: making a slide [1.1.1–2]



1. Start with the **low-power** objective (×10); focus with the **coarse** knob.
2. Switch to **high power** (×40); focus with the **fine** knob only.
3. Total magnification = eyepiece × objective, e.g. 10 × 40 = **×400**.

Onion epidermis (plant cells)

1. Peel the thin inner skin from a piece of onion with forceps.
2. Lay it **flat** on a slide in a drop of water — no folds.
3. Add a drop of **iodine solution** (stains the nucleus and any starch).
4. Lower a coverslip at an angle with a mounted needle to **avoid air bubbles**.

Cheek cells (animal cells)

1. Gently scrape the inside of your cheek with a **clean** cotton bud.
2. Smear it on a slide; add **methylene blue** (stains the nucleus blue).
3. Coverslip on at an angle; blot extra stain; dispose of the bud in disinfectant.

why stain? 🎨
Cells are almost **transparent**. A stain is absorbed by some parts (like the nucleus) more than others, giving **contrast** so you can see them.

✱ What you actually see

Onion cells look like **brick-shaped** boxes in neat rows, each with a dark nucleus near the wall — but **no chloroplasts** (onion bulbs grow underground). Cheek cells look like flattened, **irregular** blobs with a blue nucleus in the middle, often folded or overlapping.

	light microscope	*electron microscope*
uses	light; living or stained cells	a beam of electrons
shows	nucleus, cytoplasm, wall, chloroplasts	also **ribosomes**, mitochondria in detail
image	photomicrograph	electron micrograph

✱ Levels of organisation [1.2.2]

cell	tissue	organ	organ system	organism
basic unit of life	similar cells together	several tissues	organs working together	a whole living thing
muscle cell	muscle tissue	heart	circulatory system	human
palisade cell	palisade tissue	leaf	shoot system	sunflower

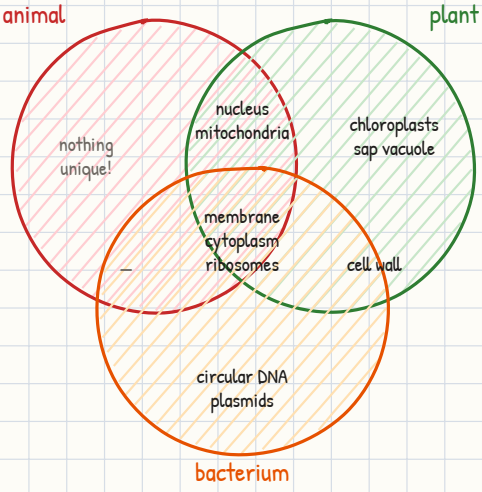
animal example in maroon · plant example in green

tissue: a group of **similar cells** working together to do a shared function.
organ: a structure made of **several tissues** working together to do specific functions.
organ system: a group of **organs** with related functions working together.

Examples you should know: the **digestive system** (organs: stomach, intestines, liver…), the **circulatory system** (heart, blood vessels), the **respiratory system** (trachea, lungs). In plants: a **leaf** is an organ made of epidermis, palisade, spongy mesophyll and vascular tissues.

record breakers 🏆
The longest human cell is a **neurone** running from your spine to your big toe — about 1 m long! The biggest single cell is an **ostrich egg** yolk.

✱ Who has what? (a Venn diagram)



Preview

1.2 Specialised cells, tissues and organs

Cells become **specialised**: their structure is adapted to their function. [1.2.1] Learn the feature **and** say why it helps.

drawing	cell (where)	special features ⇒ why they help
	**ciliated cell** lining trachea & bronchi	tiny hairs (**cilia**) that beat ⇒ sweep **mucus** with trapped dust and pathogens up and away from the lungs
	**root hair cell** root surface	long **hair-like extension** ⇒ large **surface area** for absorbing water (osmosis) and mineral ions (active transport)
	**palisade mesophyll cell** upper part of leaf	packed with **chloroplasts**; tall, column shape so many fit under the upper surface ⇒ absorbs lots of light for **photosynthesis**
	**neurone** nerves, brain, spinal cord	very **long** fibre (axon) ⇒ carries **electrical impulses** over long distances; branched endings connect to many cells
	**red blood cell** blood	full of **haemoglobin**; **no nucleus** (more room); **biconcave** disc (large surface area) ⇒ carries **oxygen**; flexible to squeeze through capillaries
	**sperm cell** testes	**flagellum** to swim · many **mitochondria** to release energy · **acrosome** with enzymes to digest into the egg
	**egg cell** ovaries	large, with **energy stores** in the cytoplasm for the early embryo · **jelly coat** changes after fertilisation so only one sperm gets in

✱ Magnification [1.2.3]

$$\text{magnification} = \frac{\text{image size}}{\text{actual size}}$$

no units · image and actual size in the **same** unit



Rearrange it like any triangle: **actual = image ÷ magnification** · **image = actual × magnification**.

how small is small?

plant cell ≈ 50–100 µm · cheek cell ≈ 60 µm · red blood cell ≈ 7.5 µm · bacterium ≈ 1–5 µm · virus ≈ 0.1 µm. You could line up about **130 red blood cells** across 1 mm!

WE 1 ✎ find the magnification

A drawing of a pollen grain is 45 mm wide. The real grain is 0.15 mm wide.

magnification = 45 ÷ 0.15 = **×300**

WE 2 ✎ find the actual size (in µm)

A cheek cell measures 36 mm in a photomicrograph at ×600.

actual = 36 ÷ 600 = 0.06 mm

0.06 × 1000 = **60 µm**

WE 3 ✎ measure it yourself

A photo shows a red blood cell 30 mm across; real size 7.5 µm.

convert first: 30 mm = 30 000 µm

magnification = 30 000 ÷ 7.5 = **×4000**

top error ⚠ dividing mm by µm. Convert to the **same unit** first!

Preview



Topic 1 · cell-f test

Practice Questions



PAPER 1 · MULTIPLE CHOICE

1 Which structure is found in plant cells but **not** in animal cells?

A cell membrane B cytoplasm
C cellulose cell wall D mitochondrion

2 What is the function of ribosomes?

A protein synthesis
B aerobic respiration
C photosynthesis
D controlling what enters the cell

3 In which structure does aerobic respiration take place?

A nucleus B ribosome C mitochondrion D chloroplast

4 Which **pair** of structures is found in a bacterial cell?

A nucleus and plasmids
B circular DNA and plasmids
C mitochondria and cell wall
D chloroplasts and ribosomes

5 Which structure controls the movement of substances into and out of a cell?

A cell wall B cell membrane
C cytoplasm D sap vacuole

6 Which structure is **not** found in a root hair cell?

A cell wall B chloroplast C nucleus D vacuole

7 Which stain is used to make the nuclei of cheek cells visible?

A iodine solution B Benedict's solution
C methylene blue D biuret solution

8 Which sequence is in order of increasing complexity?

A cell → organ → tissue → organ system
B tissue → cell → organ → organ system
C cell → tissue → organ → organ system
D organ → tissue → cell → organism

9 What is an organ?

A a group of similar cells with the same function
B a group of tissues working together to do specific functions
C a group of organs working together
D a single cell with a special function

10 Which cell has no nucleus and a biconcave shape?

A egg cell B neurone
C red blood cell D palisade cell

11 Which part of a sperm cell releases energy for swimming?

A acrosome B flagellum C mitochondria D nucleus

12 A cell is 0.02 mm long. A drawing of it is 40 mm long. What is the magnification?

A ×20 B ×200 C ×800 D ×2000

13 A photograph of a cell at ×500 is 15 mm wide. How wide is the real cell?

A 3 µm B 30 µm C 300 µm D 7500 µm

14 Which structure is present in animal, plant **and** bacterial cells?

A nucleus B ribosome
C chloroplast D mitochondrion

15 What is the main function of the sap vacuole?

A to make proteins
B to absorb light
C to store cell sap and keep the cell turgid
D to release energy from glucose

16 A microscope has a ×10 eyepiece and a ×40 objective. What is the total magnification?

A ×4 B ×50 C ×400 D ×4000

Warm-up: name the structure *(answers at the back)*

1. contains chlorophyll
2. site of protein synthesis
3. small rings of DNA in bacteria
4. partially permeable layer
5. made of cellulose
6. contains the chromosomes

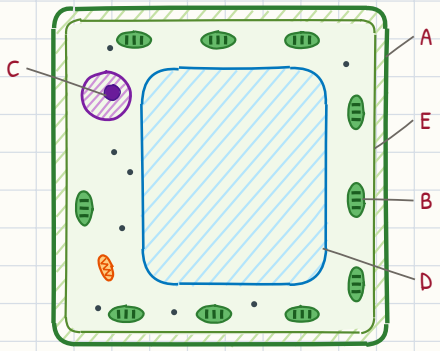
MCQ trick

In magnification questions, write the units next to **every** number before you divide. If one is in mm and the other in µm, multiply the mm value by 1000 first. Wrong options are often the "forgot to convert" answers — check the powers of 1000!

Preview

PAPER 2 · STRUCTURED

1 The diagram shows a palisade mesophyll cell.



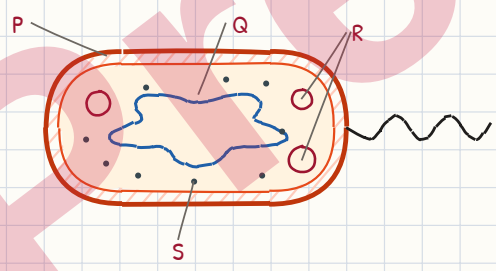
(a) Name the structures labelled A to E. [5]

(b) State the function of B and of C. [2]

(c) Name **two** structures, not labelled, that are found in this cell and in animal cells. [2]

(d) A root hair cell does not have structure B. Suggest why. [1]

2 The diagram shows a bacterial cell.



(a) Name P, Q, R and S. [4]

(b) State **two** ways in which the structure of this cell is different from the structure of an animal cell. [2]

3 A student prepares a slide of cheek cells.

(a) Describe how the student should prepare the slide. [3]

(b) Explain why a stain is used. [1]

(c) The student draws a cheek cell 54 mm wide. The real cell is 0.06 mm wide. Calculate the magnification of the drawing. [2]

(d) In a photomicrograph at ×800, a nucleus is 8 mm wide. Calculate the actual width of the nucleus in µm. Show your working. [3]

4 Cells, tissues and organs.

(a) Explain how **two** features of a red blood cell are related to its function. [4]

(b) State how a ciliated cell protects the lungs. [2]

(c) Explain the difference between a tissue and an organ. [2]

(d) Name the organ system that contains the heart. [1]

Preview

PREVIEW

Topic 2

Classification

sorting the living world into groups

2.1 Concept and use of a classification system

There are millions of species, so biologists sort organisms into groups using the **features they share**. [2.1.1] Organisms in the same group usually share a common ancestor — so classification also shows how closely related they are.

> **species**: a group of organisms that can reproduce to produce fertile offspring. [2.1.2]

A horse and a donkey can mate, but their offspring (a **mule**) is **infertile** — so horses and donkeys are **different species**.

✱ The binomial system [2.1.3]

An **internationally agreed** system: every species gets a two-part scientific name — **genus** + **species**.

common name	genus	species
human	*Homo*	*sapiens*
lion	*Panthera*	*leo*
tiger	*Panthera*	*tigris*
rice	*Oryza*	*sativa*

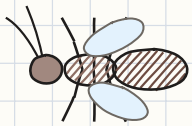
Rules: genus with a **capital**, species **lower case**; *italics* when typed, underlined when handwritten. Lions and tigers share a genus ⇒ close relatives.

why not just say "lion"?

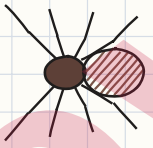
A "robin" in Britain and a "robin" in the USA are totally different birds! Scientific names are the same in every language and every country, so scientists everywhere know exactly which organism is meant.

✱ Dichotomous keys [2.1.4]

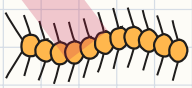
Dichotomous = "splitting in two". Each step asks about one visible feature with only two possible answers; follow the answers until you reach a name.



A



B



C

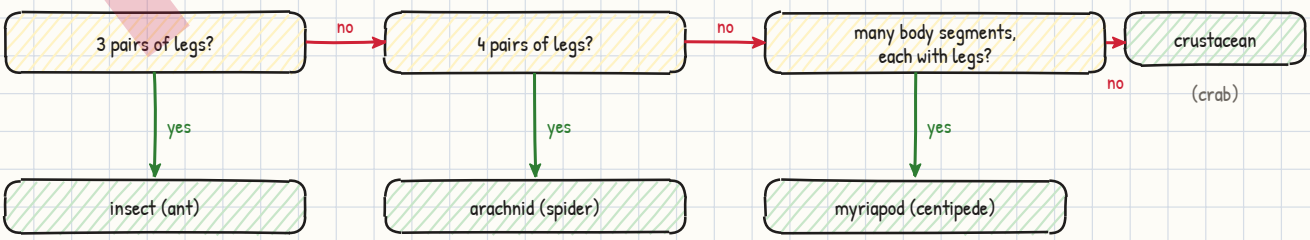


D

step	feature	go to
1a	three pairs of legs	**insect** (A)
1b	more than three pairs of legs	go to 2
2a	four pairs of legs	**arachnid** (B)
2b	more than four pairs of legs	go to 3
3a	many body segments, each with legs	**myriapod** (C)
3b	five pairs of legs, first pair are claws	**crustacean** (D)

Making your own key — full marks every time:

1. choose a feature that splits the group into **two**;
2. use features you can **see** in the drawing (legs, wings, shape) — not colour, size or "lives in water";
3. write each step as an **opposite pair** (a/b), e.g. "has wings / has no wings";
4. keep going until **each organism** ends up alone. For *n* organisms you need ***n* − 1 steps**.



the same key drawn as a branching (flowchart) key

Preview

2.2 Features of organisms

✱ The five kingdoms [2.2.1]



animal	plant	fungus	prokaryote	protoctist
multicellular	multicellular	usually multicellular (thread-like **hyphae**); yeast is single-celled	**single-celled**	mostly **single-celled**
cells have a nucleus; **no cell wall**	nucleus; **cellulose** cell walls	nucleus; cell walls of **chitin** (not cellulose)	**no nucleus**; circular DNA, plasmids; cell wall (not cellulose)	cells **with a nucleus**
no chlorophyll: feed on organic substances made by other organisms	**chloroplasts**: make their own food by **photosynthesis**	no chlorophyll: secrete enzymes onto food and **absorb** the products (decomposers)	most feed on other organisms; a few photosynthesise	some **animal-like** (*Amoeba*, *Plasmodium*), some **plant-like** with chloroplasts (algae)
nervous coordination; often move about	store **starch** or sucrose	store **glycogen**	tiny: about 1–5 µm	the "doesn't fit anywhere else" group
e.g. insects, fish, humans	e.g. ferns, maize, mango tree	e.g. mushroom, bread mould, yeast	e.g. bacteria such as *E. coli*	e.g. *Amoeba*, *Chlorella*, seaweeds

✱ The animal kingdom: vertebrates (animals with a backbone) [2.2.2a]

group	body covering	breathing	reproduction	other features
mammals	**fur / hair**	lungs	most give birth to live young; **placenta**	**mammary glands** (milk); external ears (**pinna**); whiskers; different types of teeth
birds	**feathers** (scales on legs)	lungs	eggs with **hard shells** laid on land	**beak**; front limbs are **wings**
reptiles	**dry, scaly** skin	lungs	eggs with **rubbery / leathery shells** laid on land	e.g. snakes, lizards, crocodiles, turtles
amphibians	**moist**, smooth skin, **no scales**	lungs and skin (larvae: **gills**)	jelly-coated eggs laid in **water**; tadpole larvae	adults live on land and in water, e.g. frogs, toads
fish	wet **scales**	**gills**	eggs without shells laid in water	**fins**; streamlined; lateral line

is a whale a fish?

No! It lives in water and has fins, but it has **hair** (a few bristles), breathes with **lungs**, gives birth to **live young** and feeds them **milk** – it's a **mammal**. Always classify by the key **features**, not by where it lives.

exam favourite "State **two** features visible in the photo that show this animal is a bird / reptile..." – choose features from the table that you can actually **see** (feathers, beak, scales, fins, ears), not internal ones (lungs, placenta).

memory hook Fish Amphibians Reptiles Birds Mammals – "Frogs Are Really Big Monsters" (in order of moving onto land).

✱ Classifying an organism [2.2.4]

WE 1 ✎ which group?

Bat: has fur, gives birth to live young, flies with wings. ⇒ fur + live young + milk = mammal (wings don't make it a bird – no feathers!)

Newt: moist smooth skin, no scales, lays jelly-coated eggs in a pond. ⇒ amphibian

Tortoise: dry scaly skin, lays eggs with leathery shells on land. ⇒ reptile

WE 2 ✎ arthropod or not?

Woodlouse: 7 pairs of jointed legs, 2 pairs of antennae, exoskeleton ⇒ crustacean (it lives on land, but the legs and antennae decide)

Earthworm: segmented, but **no legs** and no exoskeleton ⇒ not an arthropod

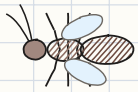
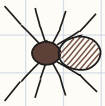
Tick: 4 pairs of legs, no antennae ⇒ arachnid

Preview

PREVIEW

✱ Arthropods: animals with jointed legs [2.2.2b]

All arthropods have an **exoskeleton** (hard outer skeleton), a **segmented body** and **jointed legs**. Count the **legs**, **body sections** and **antennae**:

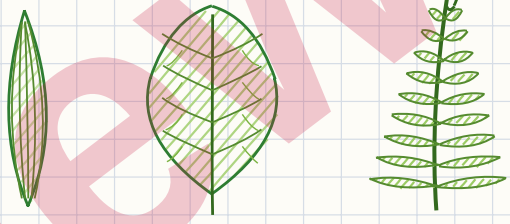
group	drawing	legs	body	antennae	other
insects		**3 pairs** (6)	**3 parts**: head, thorax, abdomen	**1 pair**	usually **2 pairs of wings**; e.g. ant, bee, butterfly
arachnids		**4 pairs** (8)	**2 parts**: cephalothorax, abdomen	**none**	no wings; e.g. spider, scorpion, tick
crustaceans		**more than 4 pairs** (often 5)	head and thorax often joined; chalky exoskeleton	**2 pairs**	most live in water; e.g. crab, shrimp, woodlouse
myriapods		**many**: 1 or 2 pairs on every segment	**many** similar segments	**1 pair**	e.g. centipede (1 pair per segment), millipede (2 pairs)

✱ The plant kingdom [2.2.3]

	ferns	flowering plants
leaves	large divided leaves called **fronds** (curled when young)	leaves of many shapes
reproduce by	**spores** (made under the fronds) — **no flowers, no seeds**	**flowers** ⟶ **seeds** inside **fruits**
transport	both have roots, stems and xylem & phloem	

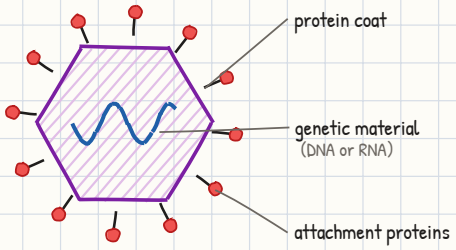
Flowering plants split into two groups:

	monocotyledons	dicotyledons
seed	**one** cotyledon (seed leaf)	**two** cotyledons
leaf veins	**parallel**; long narrow leaves	**branching network**; broad leaves
flower parts	in **threes** (3, 6, 9…)	in **fours or fives**
examples	grasses, maize, rice, wheat, lily	bean, rose, sunflower, mango



monocot leaf *dicot leaf* *fern frond*

✱ Viruses [2.2.5–6]



- just a **protein coat** around some **genetic material** — no cytoplasm, membrane, ribosomes or nucleus
- far smaller than bacteria (about 0.1 µm)
- can **only reproduce inside living cells**: they take over the host cell to make new viruses, damaging or killing it
- not placed in any of the five kingdoms

alive or not? 🤔

Outside a cell a virus does nothing at all — no feeding, no respiration, no growth. It only "comes alive" when it gets inside a host cell. That's why scientists argue about whether viruses are living things!

Topic 2 summary ✦

1. Species: can breed to give **fertile** offspring.
2. Binomial name = *Genus species*.
3. Keys: two choices per step, **visible** features.
4. 5 kingdoms: animal, plant, fungus, prokaryote, protoctist.
5. Vertebrates: fur, feathers, dry scales, moist skin, wet scales + gills.
6. Insects 6 legs, arachnids 8; viruses = coat + genes, only replicate in cells.

Preview

Topic 2 · sort it out

Practice Questions

PAPER 1 · MULTIPLE CHOICE

1 What is a species?

A organisms that look the same
B organisms that live in the same habitat
C a group of organisms that can reproduce to produce fertile offspring
D organisms in the same kingdom

2 The scientific name of the domestic cat is *Felis catus*. What does *Felis* show?

A the species B the genus C the kingdom D the family

3 Which animal has fur, external ears and mammary glands?

A bird B reptile C mammal D amphibian

4 An animal has moist skin without scales and lays its eggs in water. What is it?

A an amphibian B a fish
C a reptile D a mammal

5 Which feature is found in birds but in no other vertebrate group?

A lungs B feathers
C hard-shelled eggs D a backbone

6 An arthropod has four pairs of legs and two body sections. Which group is it in?

A insects B crustaceans C arachnids D myriapods

7 How many pairs of antennae does a crustacean have?

A none B one C two D three

8 Which kingdom contains organisms with cell walls made of chitin that feed by absorbing digested food?

A plant B prokaryote C protoctist D fungus

9 Which kingdom contains organisms whose cells have **no** nucleus?

A animal B fungus C prokaryote D protoctist

10 A plant has leaves with parallel veins and flower parts in threes. What is it?

A a fern B a monocotyledon
C a dicotyledon D a fungus

11 Ferns are different from flowering plants because ferns

A have no roots
B do not photosynthesise
C reproduce by spores, not seeds
D have no xylem

12 What is a virus made of?

A cytoplasm and a cell wall
B a protein coat and genetic material
C a nucleus and a membrane
D ribosomes and DNA

13 Why are viruses not grown on agar jelly like bacteria?

A they are too large
B they can only reproduce inside living cells
C they need light
D agar kills them

14 Which two animals are most closely related? *Canis lupus*, *Canis latrans*, *Vulpes vulpes*, *Felis catus*

A *Canis lupus* and *Vulpes vulpes*
B *Canis lupus* and *Canis latrans*
C *Vulpes vulpes* and *Felis catus*
D *Canis latrans* and *Felis catus*

15 A key is used to identify 6 organisms. What is the smallest number of steps the key needs?

A 3 B 5 C 6 D 12

16 Which organism is a protoctist?

A yeast B *Amoeba* C *E. coli* D a fern

Warm-up: which vertebrate group? (answers at the back)

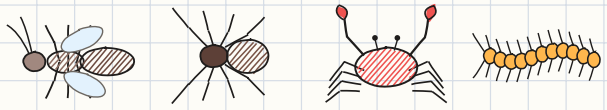
1. shark
2. crocodile
3. penguin
4. dolphin
5. frog
6. bat

key-question trick 🎯
Work through a key with your pencil, writing the step number next to each drawing as you go. Never skip a step — the question writers put "tempting" organisms that look right at step 1 but split off at step 3!

Preview

PAPER 2 · STRUCTURED

1 The drawings show four arthropods (not to scale).



E F G H

(a) State **two** features shared by all arthropods. [2]

(b) Construct a dichotomous key to identify E, F, G and H, using features visible in the drawings. [4]

(c) Name the arthropod group of F and state **one** feature you used. [2]

2 The binomial system.

(a) Describe the binomial system of naming species. [2]

(b) The lion is *Panthera leo* and the leopard is *Panthera pardus*. Explain what these names show about the two animals. [2]

(c) A lion and a tiger can produce a cub called a liger, which is usually infertile. Explain why lions and tigers are classed as different species. [2]

3 Vertebrates and plants.

(a) A student finds an animal with dry scaly skin that lays eggs with leathery shells on land. Name its vertebrate group. [1]

(b) State **three** features that distinguish mammals from all other vertebrates. [3]

(c) State **two** differences between the leaves of a monocotyledon and a dicotyledon. [2]

(d) State how ferns reproduce. [1]

4 Microorganisms are classified in different ways.

(a) Name the kingdom that contains yeast and mushrooms, and give **one** feature of this kingdom. [2]

(b) Describe the structure of a virus. [2]

(c) Explain why viruses are not placed in any of the five kingdoms. [2]

(d) State **one** feature that a protoctist has but a prokaryote does not. [1]

Preview

Quick-fire recall ✦ *P1 = 16 · P2 = 28*

1. Define species; name the 2 parts of a binomial name.
2. Feature of each of the 5 vertebrate groups.
3. Legs, body parts, antennae of the 4 arthropod groups.
4. Monocot vs dicot: seed, veins, flower parts.
5. Fern vs flowering plant.
6. Virus = protein coat + genetic material; only inside cells.